

End Semester Exam

School of Mathematics and Statistics
University of Hyderabad

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Duration: 3 Hours
Maximum Score: 60 points

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Course: Eng Math - II

Instructions: You may use results proven in the lectures; however, answers without justification will receive a score of zero.

1. Let k be a number selected at random from $\{1, 2, 3, \dots, 2024\}$. What is the probability that k is divisible by 9 and 15 ? [6]
2. There are 30 True or False questions in an examination. A student knows the answer to 20 questions and guesses the answers to the remaining 10 questions at random. What is the probability that the student gets exactly 24 answers correct ? [6]
3. An elementary school is offering 3 language classes: one in Spanish, one in French, and one in German. The classes are open to any of the 100 students in the school. There are 28 students in the Spanish class, 26 in the French class. and 16 in the German class. There are 12 students who are in both Spanish and French, 4 who are in both Spanish and German, and 6 who are in both French and German. In addition, there are 2 students taking all 3 classes. If a student is chosen randomly, what is the probability that he or she is not in any of the language classes ? [6]
4. An economic model predicts that if interest rates rise, then there is a 60 percent chance that unemployment will increase, but that if interest rates do not rise, then there is only a 30 percent chance that unemployment will increase. If the economist believes there is a 40 percent chance that interest rates will rise, what is the probability that unemployment will increase ? [6]
5. Define geometric random variable and compute the expected value of a geometric random variable. [3 + 3]
6. Let $f : \mathbb{C} \setminus \{1\} \longrightarrow \mathbb{C}$ be defined by
$$f(z) = \frac{\bar{z} - 1}{z - 1}.$$
Examine whether $\lim_{z \rightarrow 1} f(z)$ exists. [6]

7. If f is analytic in a domain D and $|f(z)|$ is constant in D , then show that $f(z)$ must be constant in D . [6]

8. Find the Maclaurin series expansion of the function

$$f(z) = \frac{z}{z^2 + 9}$$

[6]

9. Using the Cauchy's Residue Theorem, evaluate the integral

$$\int_C \frac{z+1}{z(z-2)} dz$$

where C is the circle $|z| = 3$ described in counterclockwise.

[6]

10. Find all the singular and isolated singular points of the function

$$\frac{1}{\sin \frac{\pi}{z}}.$$

[3 + 3]

All the best !